Nathan Van Utrecht

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Education

University of California San Diego — San Diego, CA
M.S. Intelligent Systems, Robotics, and Control — *Sep. 2025 – Mar. 2027*

Iowa State University — Ames, IA
B.S. Mechanical Engineering; GPA: 3.98/4.00; Summa Cum Laude — *Aug. 2021 – May 2025*

Experience

Graduate Student Researcher — Jan. 2026 – Present
Xiaolong Wang's Lab, UC San Diego — *San Diego, CA*

- Built the sim-to-real stack including calibrated-camera AprilTag cube pose tracking, domain randomization, and an async policy server; deployed on a physical Dynamixel hand that solved a scrambled Rubik's Cube
- Trained policies with JAX PPO on GPU, scaling a single cross-embodiment policy to thousands of generated hand designs, using parallel environments across multiple GPUs
- Co-designing dexterous hands and manipulation policies for Rubik's Cube solving on an Amazon-sponsored project; built MuJoCo/MJX environments and reward design for cube reorientation, flipping, and layer rotation

Software Engineering Intern — June 2026 – Sep. 2026
Brain Corp — *San Diego, CA*

- Built a barcode-constrained visual SLAM stitcher using sparse bundle adjustment (nonlinear least squares) to jointly solve robot trajectory and camera extrinsics, reaching about 1 mm accuracy on the rack and 48x faster than the existing SIFT-based stitcher
- Designed and shipped world-coordinate pallet-to-location association for a 4-camera warehouse robot, casting barcode reads as camera rays onto the rack face with WMS-aware arbitration; raised precision from 0.33 to 0.93 at 100% recall, now live at 10+ customer sites
- Built, as sole author, a data-driven evaluation platform (FastAPI, BigQuery, React) with a labeled golden set and commit-pinned pipeline runs, extended into a harness for AI coding agents

Undergraduate Researcher (Honors Capstone) — Nov. 2023 – May 2025
Coordinated Systems Lab, Iowa State University — *Ames, IA*

- Benchmarked AIRL, GAIL, and Behavioral Cloning across 4 MuJoCo environments under shifted mass, friction, actuation, morphology, and goals; AIRL beat direct imitation by over 140% on transfer tasks
- Showed a transferred AIRL reward trains a new agent to 98% of oracle performance in a modified Hopper, and identified failure modes under out-of-distribution morphology and goal changes

REU Researcher — May 2024 – Aug. 2024
Translational Artificial Intelligence Center, Iowa State University — *Ames, IA*

- Benchmarked model-free SAC and PPO against model-based SHAC for sim-to-real robustness; SAC converged 5× faster and was more robust to system noise; reimplemented environments as differentiable simulators in PyTorch

Projects

Visual-Inertial SLAM | *EKF, Sensor Fusion, Stereo Vision*

- Joint SE(3) EKF fusing IMU with tracked stereo features to estimate ego-motion on a Clearpath Jackal

LiDAR SLAM with Factor Graphs | *GTSAM, ICP, Pose Graphs*

- Encoder/IMU odometry, ICP scan matching, and GTSAM pose-graph optimization for localization and mapping; proximity loop closures cut graph error 65% vs 43% for fixed-interval

Data-Efficient Koopman Modeling with KANs | *PyTorch, Koopman Operators, Soft Robotics*

- KANs as learned Koopman observables for a soft robot arm; under half the training data of polynomial EDMD, about 2 ms per prediction

Technical Skills

Languages: Python, C/C++, TypeScript, MATLAB, SQL
Robot Learning: PyTorch, JAX, MuJoCo/MJX, Gymnasium, PPO, SAC, AIRL/GAIL, Sim-to-Real
Perception & Estimation: Bundle Adjustment, Nonlinear Least Squares, Visual SLAM, EKF, Calibration
Infrastructure: GCP (Cloud Run, BigQuery), FastAPI, React, ROS, Docker, Git, Linux